



RAFFLES INSTITUTION
2014 Year 6
Higher 3

MATHEMATICS

9824

Additional materials: List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

Answer **all** the questions.

1. The sequence $x_1, x_2, x_3, \dots$ is defined by $x_1 = x_2 = \frac{3}{2}$ and $x_n = \frac{x_{n-1}x_{n-2} + 1}{x_{n-1} + x_{n-2}}$, $n \geq 3$. Another sequence $y_1, y_2, y_3, \dots$ is defined by $y_n = x_n - 1$, $n \geq 1$.

(i) Show that $y_n = \frac{y_{n-1}y_{n-2}}{y_{n-1} + y_{n-2} + 2}$ for $n \geq 3$. [1]

(ii) Prove by mathematical induction that $0 < y_n < 1$ for $n \geq 1$. [5]

(iii) Using (i) and (ii), show that $y_n < \frac{1}{2}y_{n-1}$ for $n \geq 3$. [1]

(iv) Hence show that the series $y_1 + y_2 + y_3 + \dots$ converges. [3]

(v) The proof of the convergence of the sequence $z_n = \sqrt[n]{x_1 x_2 \dots x_n}$, $n \geq 1$ is as follows:

Let $n \geq 3$ be a positive integer. Then,

$$1 < \sqrt[n]{x_1 x_2 \dots x_n} \quad (1)$$

$$< \frac{x_1 + x_2 + \dots + x_n}{n} \quad (2)$$

$$< 1 + \frac{3}{2n} \quad (3)$$

Therefore, the sequence $z_1, z_2, z_3, \dots$ converges to 1.

Justify steps (1), (2) and (3) in the proof above. [3]

2. (i) By using the AM-GM inequality, show that $\frac{x^6}{(x^2 + 1)^4} \leq \frac{1}{16}$ for $-1 \leq x \leq 1$. [2]

(ii) Find constants A, B, C and D such that for all real numbers x ,

$$\frac{1}{(x^2 + 1)^4} = \frac{d}{dx} \left(\frac{Ax^5 + Bx^3 + Cx}{(x^2 + 1)^3} \right) + \frac{Dx^6}{(x^2 + 1)^4}.$$

[4]

(iii) Using parts (i) and (ii), show that $\frac{11}{12} \leq \int_{-1}^1 \frac{1}{(x^2 + 1)^4} dx \leq \frac{25}{24}$. [6]

3. [In this question, a function f is said to be **convex** if $f''(x) > 0$ for all x in its domain.]
Let f be a convex function.

(i) Show that f' is a one to one function. [1]

Since f' is one to one, for each p in the range of f' , there is exactly one tangent to the curve $y = f(x)$ that has gradient p . Let $g(p)$ be the negative of the y -intercept of this tangent.

(ii) Show that $g(p) = px_p - f(x_p)$, where $f'(x_p) = p$. [2]

This defines a function g , with domain the range of f' , that is called the Legendre transform of f .

(iii) Treating p as a fixed value, show that $g(p)$ is the maximum value of $xp - f(x)$. [2]

For the following parts, f is defined by $f(x) = \frac{x^m}{m}$, $x > 0$, where $m > 1$ is a real number and n is defined by $\frac{1}{m} + \frac{1}{n} = 1$.

(iv) Verify that f is a convex function and obtain a formula for its Legendre transform $g(p)$. Express your answer in terms of p and n . [4]

(v) Using the results in this question, show that $xp \leq \frac{x^m}{m} + \frac{p^n}{n}$ for all $x, p > 0$. Determine also the condition for equality to hold. [2]

4. (a) Use a combinatorial argument to show that $\sum_{k=2}^n \binom{k}{2} = \binom{n+1}{3}$. [3]

(b) A positive integer is said to be **good** if the sum of its digits is 9. The sequence $a_1, a_2, \dots$ consists of all good integers arranged in ascending order.

(i) Show that the number of n -digit **good** integers is given by $\binom{n+7}{8}$. [3]

(ii) Determine m such that a_m is the smallest 7-digit **good** integer. [1]

(iii) Find a_{2014} . [2]

(c) Let $S(r)$ denote the number of ways to distribute r distinct objects into 3 identical boxes such that no box is empty.

(i) Given that $r \geq 3$, show that $S(r) = 2^{r-2} - 1 + 3S(r-1)$. [3]

(ii) Hence determine the number of ways to distribute 2-digit **good** integers into 3 non-empty sets. [1]

5. We say a permutation $(a_1, a_2, \dots, a_{2n})$ of the numbers $1, 2, \dots, 2n$ is **great** if $|a_i - a_{i+1}| = n$ for at least one $i \in \{1, 2, \dots, 2n - 1\}$. For example, when $n = 2$, $(4, \overline{1}, \overline{3}, 2)$ and $(\overline{4}, \overline{2}, \overline{3}, 1)$ are **great** but $(1, 2, 3, 4)$ is not. Let $G(n)$ denote the number of **great** permutations of the numbers $1, 2, \dots, 2n$.

(i) Find $G(1)$ and $G(2)$. [3]

For $1 \leq k \leq n$, let X_k be the set of all permutations of $\{1, 2, \dots, 2n\}$ with k and $k + n$ in neighbouring positions.

(ii) Show that $|X_k| = 2(2n - 1)!$. [2]

(iii) For $1 \leq i_1 < i_2 < \dots < i_m \leq n$, find $|X_{i_1} \cap X_{i_2} \cap \dots \cap X_{i_m}|$ in terms of m and n . [2]

(iv) Show that for $n \geq 2$, $G(n) \geq \sum_{k=1}^n |X_k| - \sum_{1 \leq k < m \leq n} |X_k \cap X_m|$. [2]

(v) Hence show that for every positive integer n , $\frac{G(n)}{(2n)!} > \frac{1}{2}$. [3]

6. (i) Find the general solution of the differential equation

$$\frac{dy}{dx} + 2xy = x.$$

[3]

(ii) A particular solution of the differential equation

$$y \frac{dy}{dx} + xy^2 = \frac{x}{2}$$

has $y = 0$ when $x = 1$. By using the substitution $y^2 = z$, find y^2 in terms of x . [4]

(iii) A particular solution of the differential equation

$$4 \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + y = 0$$

has $y = 1$ and $\frac{dy}{dx} = -\frac{3}{2}$ when $x = 0$. Find the exact value of y when $x = \ln 4$. [6]

7. A solution to the differential equation

$$\frac{dy}{dx} = (y-1)^2$$

has $y = -1.5$ at $x = 0$.

(i) Show that the exact value of y at $x = 1$ is $\frac{2}{7}$. [4]

(ii) Use the Euler method with step size 0.5 to estimate y at $x = 1$. [2]

(iii) Use a sketch to explain why the Euler method with the step size chosen in (ii) does not provide a good approximation to the exact solution. [3]

(iv) Copy and complete the table showing the use of the improved Euler method with step size 0.5 to estimate y at $x = 1$. [3]

x	y	$(y-1)^2$	$\tilde{y}$	$\frac{\Delta y}{\Delta x}$
0	-1.5	6.25	1.625	3.3203
0.5	0.16015	0.70535		
1				

8. Spruce budworms are serious pests that destroy coniferous forests. Their population is controlled by birds predating on the insect. To model this interaction of the spruce budworm population with their primary predator, the following model is proposed.

$$\frac{dB}{dt} = RB \left(1 - \frac{B}{Q} \right) - \frac{B^2}{1+B^2}.$$

Here B is the population of the spruce budworms and the parameter R is the growth rate coefficient which we will assume is constant.

(i) Explain the significance of the parameter Q . [1]

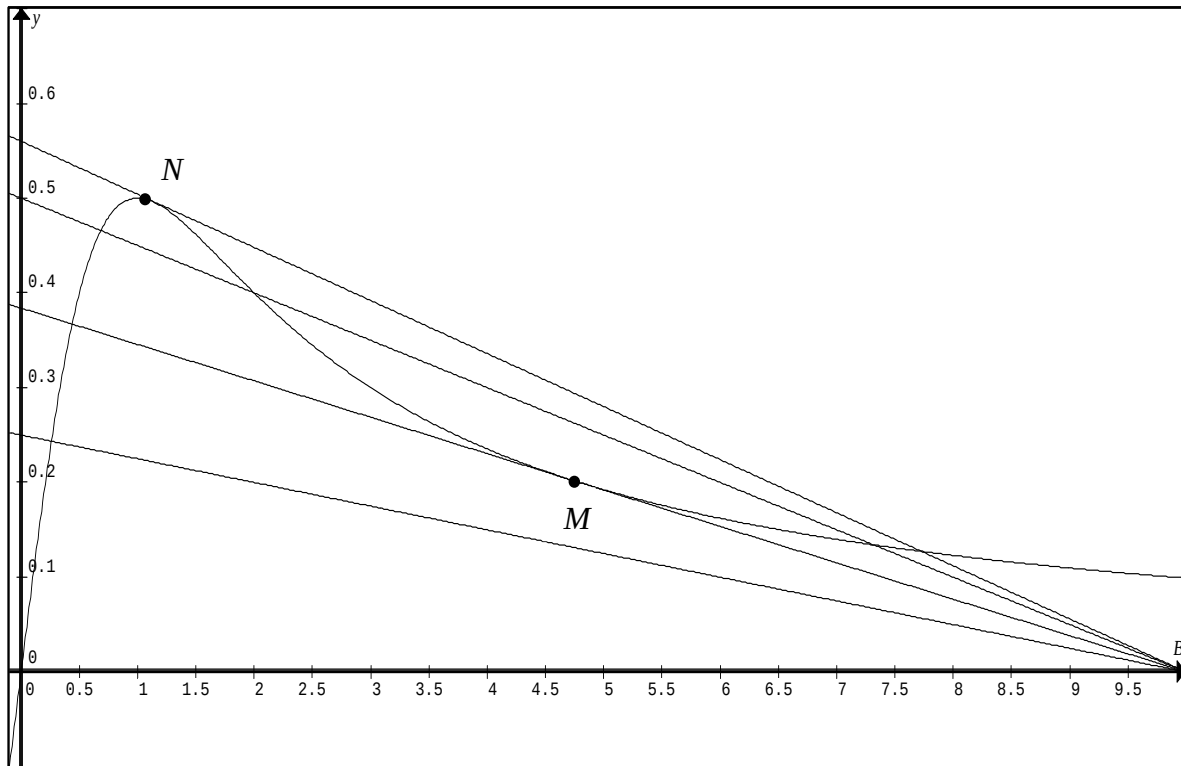
(ii) Find the series expansion of $\frac{dB}{dt}$ in ascending powers of B , up to and including the term in B . [1]

(iii) Hence show that the extinction equilibrium $B_e = 0$ is unstable. [1]

(iv) For $Q = 10$, show that the other equilibria can be found by solving the equation

$$R \left(1 - \frac{B}{10} \right) = \frac{B}{1+B^2}. \quad [1]$$

The diagram below shows the graphs of $y = \frac{B}{1+B^2}$ and $y = R\left(1 - \frac{B}{10}\right)$ for $R = 0.250, 0.384, 0.500$ and 0.560 . The lines corresponding to $R = 0.384$ and 0.560 are tangent to the graph of $y = \frac{B}{1+B^2}$ at the points M and N respectively.



- (v) Draw the phase line diagram for the case $R = 0.250$, clearly indicating the equilibrium values and comment on the stability of the equilibrium values. [4]
- (vi) Sketch a bifurcation diagram for the parameter R , showing clearly the phase lines for $R = 0, 0.250, 0.384, 0.500, 0.560$ and 0.600 . You need not indicate the equilibrium values on your diagram. [6]

[END OF PAPER]